

ME594: 3D Robotics, HW 6

Homework due on 03-02-2025, Topics: linear and angular velocity.

1. **Angular Velocity and Euler Parameters:** Consider the x-y-z convention of Euler angles, which is also known as Bryant angles. Here the rotations are parameterized by Euler angles: $\phi = 0.5$ (rotation about x-axis), $\theta = 0.25$ (rotation about y-axis), and $\psi = 0.6$ (rotation about z-axis). The euler parameters indicated the rates are $\dot{\phi}_1 = 0.1$, $\dot{\theta}_2 = 0.2$ and $\dot{\psi}_3 = 0.3$. Compute the angular velocity in the world frame ω and the angular velocity in the body frame Ω .
2. **Angular Velocity and Quaternion Derivative:** The angular velocity in world frame is given by $\omega = [0.25, -0.6, 0.3]$ and the Euler angles in the x-y-z convention are: $\phi = \pi/3$ (x-axis), $\theta = \pi/6$ (y-axis), and $\psi = \pi/2$ (z-axis). Compute the quaternion derivative and the angular velocity in body frame.

Submission: For both questions above

- **Gradescope:** Submit your work with relevant equations and numerical answers.
- **Email:** None.

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3. **Linear/Angular Velocity for the Trossen Robotics WidowX 250 S:** Figure 1 shows the

WidowX 250S by Trossen Robotics. A video of the robot in action is here:

https://www.youtube.com/watch?v=l_10IH0JNxI. I have provided the wx250s_sensors.xml files, the remaining scene and meshes can be downloaded from github https://github.com/google-deepmind/mujoco_menagerie/blob/main/trossen_wx250s/wx250s.xml

Given the following position and velocity for the first 6 joints

$q = \text{np.array}([0.5123, -1.6532, 0.5439, -0.3432, -0.1125, 0.2342])$

$u = \text{np.array}([0.2291, -0.8602, -0.8068, -0.7110, 0.0689, 0.457029])$. Compute the following:

- (a) Linear and angular velocity of the tip (red sphere).
- (b) Linear and angular velocity of the center of mass of body called “lower_forearm_link”.

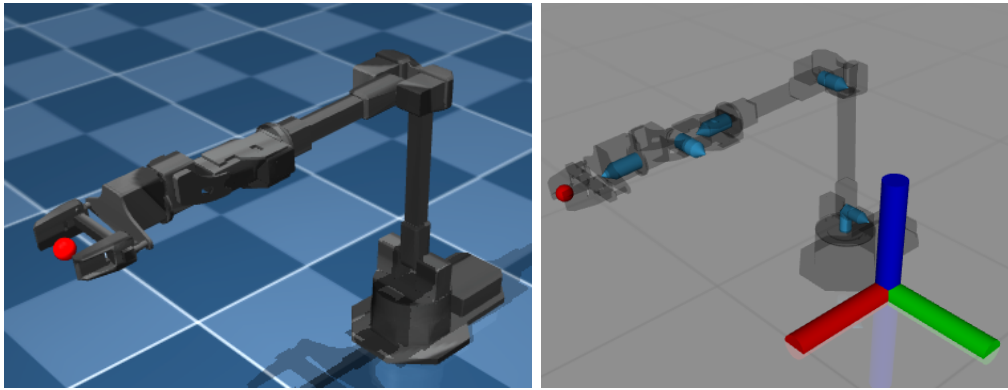


Figure 1: Perspective view of the Trossen WidowX 250 S: The robot has 6 joints, as shown in the figure on the right. The world frame is at the base of the WidowX but moved to the side in the figure to avoid clutter. The body frames are in the same direction as the world frame and centered at the joints.

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_velocity_names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “HW6 wx250s” to your subject line. If submitting as a group, please copy your group member. When the file is run, it should print the answers to the two questions.